

LAUNCESTON AP, Australia

WMO: 959660

Lat: 41.5477S

Lon: 147.2160E

Elev: 552

StdP: 14.40

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.9	31.8	25.4	19.8	41.7	27.0	21.4	40.5	29.5	51.6	26.8	50.9	4.6	190	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.6	82.7	63.7	79.0	62.6	75.7	61.0	66.4	77.1	64.8	74.7	63.2	72.5	16.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	87.6	68.7	61.0	81.8	68.0	59.2	76.7	67.0	31.4	77.4	30.1	74.7	28.9	72.4	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.7	23.2	21.0	DB	25.6	89.9	1.5	4.9	24.5	93.4	23.6	96.3	22.7	99.0	21.6	102.6	(4)
(5)			WB	25.1	69.6	1.5	1.7	24.0	70.8	23.2	71.8	22.3	72.8	21.3	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	63.0	62.8	59.4	53.5	48.9	45.3	44.4	45.7	48.4	51.8	55.7	59.4	(6)
(7)		DBStd	8.00	5.44	5.14	5.39	4.66	4.48	4.26	3.79	3.86	3.86	4.41	4.92	5.01	(7)
(8)		HDD50	681	0	0	2	19	76	150	178	143	74	31	7	1	(8)
(9)		HDD65	4436	103	94	188	345	500	592	640	599	497	411	282	185	(9)
(10)		CDD50	1823	403	359	293	124	41	9	3	9	27	85	178	292	(10)
(11)		CDD65	102	41	33	15	0	0	0	0	0	0	0	2	11	(11)
(12)		CDH74	1133	470	291	139	3	0	0	0	0	0	5	50	175	(12)
(13)		CDH80	260	132	56	28	0	0	0	0	0	0	0	6	38	(13)
(14)	Wind	WSAvg	10.2	11.3	10.4	9.8	9.1	9.2	8.6	9.4	10.4	10.8	11.1	11.1	11.4	(14)
(15)	Precipitation	PrecAvg	25.7	1.8	1.3	1.5	2.0	2.3	2.4	2.8	3.0	2.5	2.1	1.9	1.9	(15)
(16)		PrecMax	37.0	6.1	6.5	3.9	7.2	5.1	6.7	7.4	8.8	4.8	5.2	3.9	5.0	(16)
(17)		PrecMin	16.1	0.3	0.0	0.1	0.2	0.2	0.4	0.8	0.5	0.3	0.4	0.6	0.0	(17)
(18)		PrecStd	5.4	1.2	1.3	1.0	1.4	1.4	1.2	1.5	1.6	1.1	1.1	0.8	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.6	86.1	83.9	72.7	64.4	59.0	57.2	60.2	65.6	73.2	79.5	84.8	(19)
MCWB			66.8	65.3	64.4	59.5	55.8	53.3	51.0	52.1	55.0	58.3	61.8	63.2	(20)	
2%		DB	83.6	80.9	77.4	68.4	61.1	56.7	55.1	57.3	61.6	67.7	74.1	78.6	(21)	
		MCWB	63.8	64.3	62.3	57.8	54.3	51.7	49.6	50.6	52.7	56.1	59.9	61.6	(22)	
5%		DB	79.0	77.3	73.6	65.7	59.1	55.1	53.5	55.2	59.2	64.4	69.9	74.5	(23)	
		MCWB	62.3	62.5	60.3	56.1	53.3	50.9	48.7	49.3	51.5	54.6	57.9	59.5	(24)	
10%		DB	75.3	74.1	70.2	62.9	57.3	53.3	51.9	53.4	57.1	61.5	66.4	71.2	(25)	
		MCWB	60.8	60.8	58.4	54.8	52.4	49.6	48.0	48.5	50.3	52.9	55.8	58.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.0	68.4	66.5	62.1	58.0	55.5	52.7	53.9	57.0	60.5	64.7	66.2	(27)
MCD B			82.9	78.2	78.6	68.0	61.0	57.0	55.5	57.6	61.7	67.9	75.1	77.1	(28)	
2%		WB	66.3	66.5	64.1	59.7	56.4	53.1	51.2	52.2	54.7	58.0	61.7	63.8	(29)	
		MCD B	77.7	76.1	73.2	65.7	59.4	55.4	53.5	55.6	59.7	65.3	70.7	73.5	(30)	
5%		WB	64.4	64.7	62.3	58.1	54.9	51.6	50.0	50.7	52.8	56.0	59.5	61.7	(31)	
		MCD B	75.0	74.2	70.5	63.6	58.0	53.9	52.0	53.8	57.6	62.6	67.6	71.6	(32)	
10%		WB	62.5	62.8	60.4	56.3	53.3	50.3	48.8	49.3	51.2	54.0	57.5	59.5	(33)	
		MCD B	72.5	71.6	67.7	61.4	56.5	52.7	51.1	52.3	55.6	60.5	65.1	69.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.6	21.6	21.3	18.9	16.5	14.9	15.0	16.1	17.5	19.5	20.7	22.3	(35)
MCDBR			28.2	25.2	24.7	21.9	17.0	15.3	15.5	17.4	19.8	22.7	25.4	26.6	(36)	
MCWBR			12.1	11.1	11.5	12.7	11.6	11.1	11.2	11.8	12.7	13.4	13.7	12.7	(37)	
5% WB		MCDBR	24.8	22.7	22.1	19.1	14.2	13.2	13.0	14.6	18.0	20.7	22.4	24.0	(38)	
		MCWBR	12.4	11.6	11.4	12.2	10.5	10.5	10.3	10.9	12.6	12.9	13.3	13.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.331	0.323	0.312	0.302	0.289	0.289	0.305	0.325	0.331	0.333	0.333	(40)	
taud		2.502	2.542	2.556	2.563	2.567	2.590	2.599	2.550	2.483	2.475	2.494	2.508	(41)		
Ebn at Noon		314	308	296	278	259	252	261	276	291	305	313	317	(42)		
Edh at Noon		35	32	29	25	21	18	20	24	30	34	35	36	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2206	1855	1434	970	628	486	540	793	1176	1641	1971	2185	(44)	
(45)		RadStd	153	76	63	51	39	42	27	63	80	86	144	143	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.54	N/A	+0.59	+1.62	N/A	N/A	N/A	-176	+167	+36
(47) Regional (34 neighbors)		+0.58	N/A	N/A	+1.22	+0.65	+0.77	N/A	-180	+194	+25

Nomenclature: See separate page